論点2②｜一つ使っていても、もう一つを加える意味があるか

例：O→+R ＝ Oだけの予測にRを加えた改善。緑＝改善、紫＝悪化。単位：Brier skillの差 ×100。

ノード

	D→+H	D→+O	D→+R	H→+D	H→+O	H→+R	O→+D	O→+H	O→+R	R→+D	R→+H	R→+O
er_mid α=0.1	+8.8	+16.3	+0.4	-0.6	+8.6	+2.7	-0.9	+0.7	+0.2	-1.0	+10.6	+16.0
ba_mid α=0.1	+10.9	+20.2	+0.9	-0.3	+10.0	+2.4	-0.2	+0.8	+0.0	-0.2	+12.5	+19.4
amazon α=0.1	+6.5	+14.5	+3.0	+0.5	+9.5	+2.6	-0.0	+1.0	+0.5	+0.6	+6.2	+12.7
vldb α=0.1	+7.2	+23.2	+5.4	-1.7	+15.3	+3.9	-0.5	+0.5	+0.8	-1.7	+5.7	+17.3
slashdot α=0.1	+14.3	+29.3	+0.3	-0.4	+15.0	+3.3	-0.2	+0.3	+0.1	+0.7	+18.4	+29.8
enron α=0.1	+4.8	+25.0	+3.4	+6.8	+27.0	+3.1	+2.1	+2.2	+1.0	+9.9	+7.6	+30.3
er_mid α=0.01	+0.1	+0.1	+0.0	-0.0	+0.1	+0.0	+0.0	+0.1	+0.0	-0.0	+0.1	+0.1
ba_mid α=0.01	+0.1	+0.2	+0.0	+0.0	+0.2	+0.0	+0.1	+0.1	+0.0	+0.0	+0.1	+0.2
amazon α=0.01	+0.2	+0.7	+0.0	+0.2	+0.6	+0.0	+0.1	+0.0	-0.0	+0.1	+0.2	+0.8
vldb α=0.01	+0.4	+3.1	-0.0	+0.1	+3.0	+0.1	-0.1	+0.0	+0.0	+0.2	+0.5	+3.5
slashdot α=0.01	+0.2	+0.6	+0.1	-0.0	+0.4	+0.1	+0.2	+0.3	+0.1	-0.0	+0.3	+0.4
enron α=0.01	+0.4	+2.6	+0.2	+0.3	+1.8	+0.5	+0.9	+0.2	+0.3	+0.3	+0.6	+2.2

エッジ

	D→+H	D→+O	D→+R	H→+D	H→+O	H→+R	O→+D	O→+H	O→+R	R→+D	R→+H	R→+O
er_mid α=0.1	+19.9	+33.6	+18.9	+1.2	+19.0	+11.0	+0.2	+4.3	+2.5	+0.5	+11.2	+17.5
ba_mid α=0.1	+18.4	+33.6	+17.9	+3.6	+20.6	+13.0	+0.5	+2.3	+3.5	+1.9	+11.9	+20.7
amazon α=0.1	+11.6	+21.3	+11.4	+6.3	+16.4	+10.5	+1.1	+1.4	+2.2	+3.4	+7.7	+14.5
vldb α=0.1	+8.3	+27.0	+14.2	+8.9	+25.9	+17.3	+1.1	-0.6	+2.3	+2.8	+5.3	+16.8
slashdot α=0.1	+11.9	+33.4	+16.3	+12.1	+27.9	+21.1	+1.5	-4.2	+2.2	+8.0	+12.7	+25.9
enron α=0.1	+9.7	+47.3	+35.6	+18.9	+49.7	+36.4	+2.3	-4.5	+6.7	+4.1	-4.4	+20.2
er_mid α=0.01	+0.3	+0.4	+0.1	+0.9	+0.3	+0.2	+0.9	+0.1	+0.2	+0.9	+0.4	+0.5
ba_mid α=0.01	+0.4	+0.6	+0.3	+2.4	+0.6	+0.6	+2.2	+0.2	+0.4	+2.2	+0.5	+0.7
amazon α=0.01	+0.3	+1.0	+0.3	+4.5	+1.6	+1.2	+3.9	+0.2	+0.8	+3.8	+0.5	+1.4
vldb α=0.01	+0.3	+3.1	+0.7	+11.9	+7.6	+4.1	+7.5	+0.4	+2.0	+8.7	+0.6	+5.7
slashdot α=0.01	+1.5	+4.3	+3.5	+8.4	+6.6	+8.6	+7.0	+2.3	+7.5	+4.6	+2.7	+5.9
enron α=0.01	+1.8	+13.8	+4.6	+13.2	+19.0	+9.7	+6.1	-0.0	+5.1	+8.7	+2.4	+16.9



アクセス単位。初回（R未定義）と末尾100アクセスは除外。各セルは8始点の平均。ばらつきはsummary.csvのsd。
7始点で学習し、未使用の1始点で評価。目的は『次の100アクセス以内の再訪』。
すべての組み合わせを両方向で比較。A→+BとB→+Aは意味が異なる。
改善はこの予測モデルでの追加価値であり、因果・最適重み・キャッシュヒット率の改善ではない。